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Anisotropic Hall Coefficients in n-Type Germanium

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Precise measurements of Hall effect have been made at 77°K, 90°K, 195°K, and 298°K up to 11,000 gauss on oriented single crystals of n-type germanium containing 5×10^{14} antimony per cm^3 as donor centers. The results should principally indicate the variations of μ_H/μ with the magnetic field and with the temperature, since the carrier concentration may be practically constant under the present conditions.

Absolute values of μ_H/μ have been deduced from R_∞ and R_0. R_∞ for which μ_H/μ always equals unity is fortunately obtained by extraporating $R_H^{\langle 001\rangle}$(77°K) vs $1/H^2$ data. Then $R_H^{\langle 001\rangle}/R_\infty$ will straightway give $\mu_H/\mu \cdot \langle 001\rangle_H$. On the other hand, μ_H/μ for other orientations may be indirectly estimated from their extraporated R_0 for which μ_H/μ is independent of the orientation and should be equal to $R_0^{\langle 001\rangle}/R_\infty$ at respective temperatures, neglecting the effect due to slightly different impurity concentration between specimens.

Values of μ_H/μ thus determined are always smaller than unity, while those of $\langle 001\rangle_H$ approach to unity most rapidly with increasing field and those of $\langle 110\rangle_H$ most slowly. They are of course temperature dependent and the zero magnetic field values are equal to 0.83 at 77°K, 0.84 at 90°K, 0.90 at 195°K and 0.92 at 298°K respectively.

Numerical evaluation of the Boltzmann's equation taking account of ellipsoid model and mixed scatterings can give the observed values fairly well. It appears, however, preferable to introduce some correction term, e.g., reasonable amount of optical phonon scattering for the best agreement.

§ 1. Introduction

Hall measurements are the most popular and fundamental techniques to obtain many information about electronic properties in semiconductor physics. In most cases thus far, however, little attention has been paid to μ_H/μ appearing in the Hall coefficient, sometimes assuming it equals unity while sometimes equating it to $3\pi/8$ without considering scattering mechanisms in detail.

According to theoretical calculations, on the other hand, it is generally predicted that μ_H/μ is more or less dependent on the applied magnetic field and not a unique constant unless free carriers are perfectly degenerated. Usually, the behavior is rather complicated even if the semiconductor material is isotropic. For instance, Johnson and Whitesell[9] have indicated interesting results in which magnetic field dependence of μ_H/μ is consider-

ably influenced by mixing a small amount of ionized impurity scattering into the acoustic phonon scattering in the case of isotropic semiconductor, although no experimental verifications have been given because of the lack of proper isotropic materials. Theoretical analysis of anisotropic Hall coefficients in n-type germanium has been briefly dealt with by Abeles and Meiboom[2] and also by Shibuya[3] who have aimed to explain anisotropic magnetoresistance data[4] by assuming ellipsoid band model and isotropic thermal scattering. Putting the effective mass ratio at 20, they obtained 0.93 for the values of μ_H/μ at zero magnetic field. As compared with the calculations by Johnson et al, the only regreatful point in their analysis is the neglection of slight impurity scattering in the real crystals, which may considerably affect the field dependence of Hall coefficient.

As for the experimental examinations of μ_H/μ, there have been little works chiefly because of the requirements for accurate instrumentations detecting rather small effects. It is noteworthy that anisotropic variations of relative μ_H/μ values with magnetic field strengths were first observed in n-type germanium at room temperatures by Mason, Hewitt and Wick[5] and ingeniously analysed on the phenomenolygical basis of cubic symmetry only. Though their beautiful results can elucidate the relation between crystal direction and Hall coefficient, absolute values of μ_H/μ and its temperature effect have not been given and remained obscure. A drastic determination of μ_H/μ in germanium was reported by Morin[6] who compared the conductivity mobility, which was calibrated by the drift mobility of minority carriers injected in p- or n-type specimens at 300°K, with the Hall mobility of respective type. He has quoted an experimental value of 1.05 for n-type independent of temperature between 25 and 300°K. In his case, however, despite of epoch-making method, observed data were too dispresed to discuss the agreement with theory and moreover no anisotropic characteristics were considered. His brilliant success was a disclosure of extraordinary behavior of μ_H/μ in p-type germanium which will not be referred here.

Since the ellipsoid model for n-type germanium has been well established by cyclotron resonance experiments[7], theoretical investigations of transport phenomena thereafter have been pointed to the anisotropy in the relaxation time. General aspect on such a basis has been discussed by Herring and Vogt[8]. Under these circumstances, reexamination of the Hall effect as well as other transport phenomena taking account of crystal orientation and scattering anisotropy was a desirable subject for many valley semiconductors. Bullis and Krag[9] have reported a timely work on n-type germanium specimens of highly symmetric orientations, in which they have shown that observed data for μ_H/μ at 77°K agree reasonably well with calculations using ellipsoid model and constant mean free time hypothesis, especially the agreement is good when the magnetic field is in ⟨001⟩ direction and exceeds several kilo-gauss. In the second paper[10], Bullis has extended his measurements including other galvanomagnetic effects to another temperature of 300°K in addition to 77°K in order to evaluate the anisotropy factor principally due to the relaxation, where he has quoted the behaviors of Hall coefficients could not be quantitatively explained. Though these investigations have almost clarified qualitative features of μ_H/μ, there is some further information yet to be desired for better understandings.

The authors have carried out similar measurements on specimens of several selected crystal orientations at four different temperatures in the exhaustion region, and obtained the explicit variations of anisotropic μ_H/μ with temperatures and magnetic field strengths making an attempt at the highest accuracy. The present paper will describe those experimental results and the theoretical calculations aiming more quantitative approach without neglecting anisotropic factors both in effective mass and in mixed scatterings.

§ 2. Experimental Procedures

The techniques used here are almost similar to those by Bullis and Krag[9] and will be described briefly. Highly uniform and stable magnetic field up to 11,000 gauss was produced by an electromagnet for high resolution nuclear magnetic resonance of which pole face diameter was 30 cm and gap was 4.5 cm. The field strengths were determined within

0.01% errors by measuring proton resonance frequencies with a heterodyne frequency meter. The measurements of Hall voltage and sample current were carried out by a YEW Type P-7 potentiometer, where the observed values were reproducible at most within ±0.1%. Therefore, the whole accuracy was limited only by potential measurements.

Four measuring temperatures chosen here were 77°, 90°, 195°, and 298°K which were respectively provided by liquid nitrogen, liquid oxygen, mixture of dry ice and ethyl ether, and air-conditioned room temperature. In all cases the sample mounted on a holder was directly immersed in a Dewar vessel containing each refrigerant or no refrigerant.

The specimens were zone-leveled n-type germanium single crystals of about 3 ohm-cm containing nearly 5×10^{14} antimony per unit volume. Since these crystals are so well controlled that the concentrations of uncontrolled centers less than 10^{13} may be easily realizable, one may suppose an essential condition for the present purpose that the carrier concentrations are practically constant irrespective of temperatures even in the worst case. After the crystals were oriented by X-ray analysis, the specimens were cut out of the crystals in the shape as shown in Fig. 1 by a supersonic dice. The selected orientations are also indicated in Fig. 1. Angle θ varying from 0° to 90° was chosen as a parameter. In the following, a notation $R_H^{\langle\theta\rangle}(T°\mathrm{K})$ will represent the Hall coefficient at magnetic field strength of H gauss and at temperature of T°K in a specimen for which magnetic field is applied in the direction of angle θ. In place of θ, Miller indices will be used in the case of high symmetry orientations, for example, $\langle 0^\circ\rangle$ will be represented by $\langle 001\rangle$ and so on. As either sample current or Hall voltage is in $\langle 1\bar{1}0\rangle$ direction in the present case and both are permutable each other[4], indication of the direction is omitted in the notation. Although no cross magneto-resistance effect[4] appears in these orientations, Hall coefficient is determined according to the usual way by reversing the magnetic field and the sample current and averaging the observed Hall voltages. It was found in our case that the reversal of current had no important effect upon the results. First of all, the vital point for precise experiments was to

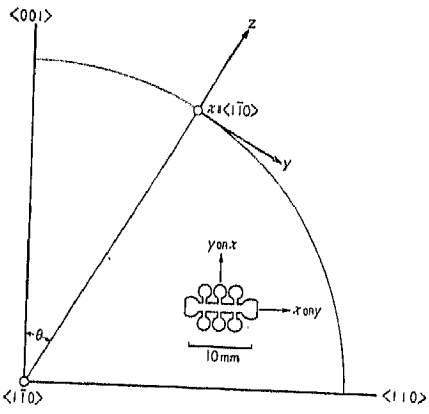


Fig. 1. Geometry of specimen and direction of magnetic field H, sample current I and Hall voltage V_H. θ is chosen as a parameter.

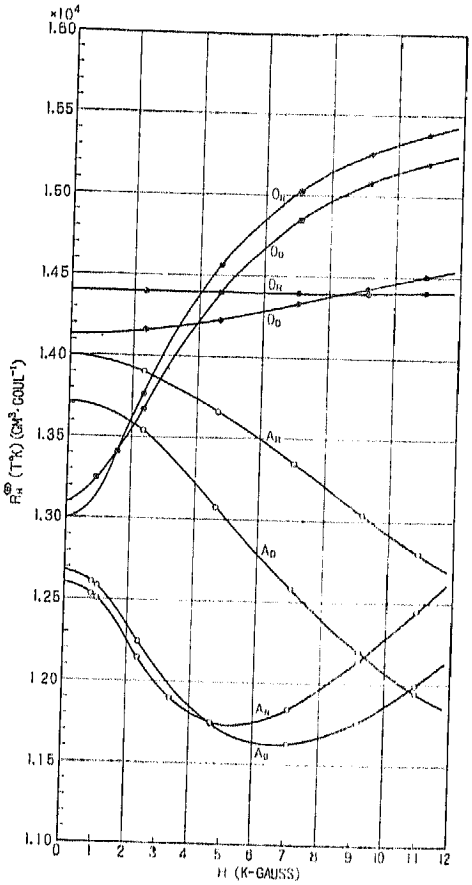


Fig. 2. $R_H^{\langle 001\rangle}(T°\mathrm{K})$ and $R_H^{\langle 110\rangle}(T°\mathrm{K})$ plotted against H. <001> sample is designated as "O", and <110> as "A". Subscripts R, D, etc. represent specified temperatures: i.e., R≡room temperature (298°K), D≡dry ice (195°K), O≡liquid oxygen (90°K), and N≡liquid nitrogen (77°K).

obtain low-resistance contacts for the specimens. In order to realize good ohmic contacts, electrode- and probe-arms were electroplated with rhodium or alloyed with tin-antimony alloy so as to be more n-type before soldering. There was no difference at room temperature between both treatments, where the contacts were always satisfactory, while at low temperatures the latter showed better results in contrast with facts that the former sometimes cracked or frequently made potentiometer balance insensible with slight contact resistance. Hence the alloying contacts were used in most specimens, though they required somewhat elaborate processes.

§ 3. Experimental Results

A. $R_H^{\langle 0\rangle}$ vs H at several specified temperatures

Several curves of Hall coefficients for two typical high symmetry orientations are shown in Fig. 2. The following facts are notable here;

(1) that both $R_0^{\langle 001\rangle}(T°\mathrm{K})$ and $R_0^{\langle 110\rangle}(T°\mathrm{K})$ increase with increasing temperatures in the same proportion, and

(2) that $R_H^{\langle 00\rangle}(298°\mathrm{K})$ is strikingly field insensitive while other R_H's in both orientations vary in quite different manners.

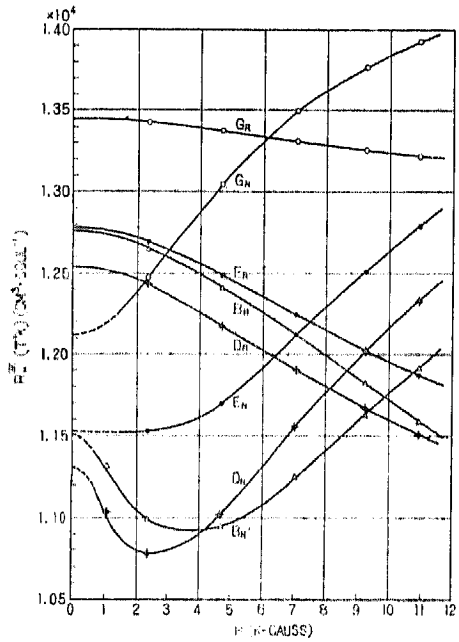


Fig. 3. $R_H^{\langle 0\rangle}(T°\mathrm{K})$ vs H for several orientations. $\langle 0\rangle=\langle 75°\rangle$ for sample B, $\langle 111\rangle$ for D, $\langle 112\rangle$ for E, and $\langle 15°\rangle$ for G. Subscripts are the same to Fig. 2.

Since carrier concentrations never decrease with increasing temperatures or hardly depend upon magnetic field strengths and directions, it is evident that behaviors of these curves should be ascribed to relative changes in μ_H/μ.

Fig. 3 shows several sets of Hall coefficients for specimens of other orientations.

B. Determination of $R_\infty^{\langle 001\rangle}(77°\mathrm{K})$ and $\mu_H/\mu \cdot\langle 001\rangle_H$

At high magnetic field and low temperatures a constant mean free time approximation for $R_H^{\langle 001\rangle}$ is very good as demonstrated by Bullis et al[9], and the Hall coefficient for an inifinite magnetic field, $R_\infty^{\langle 001\rangle}(77°\mathrm{K})$, is determined by plotting $R_H^{\langle 001\rangle}(77°\mathrm{K})$ as a function of $1/H^2$ and extraporating linearly to the value of $H=\infty$ as shown in Fig. 4. Since μ_H/μ at infinite magnetic field always equals unity irrespective of the crystal orientations,

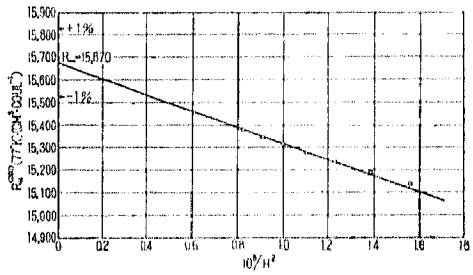


Fig. 4. $R_H^{\langle 001\rangle}(77°\mathrm{K})$ vs $1/H^2$. Determination of R_∞.

the mass ratio and the scattering mechanism as discussed in the theory. $R_H^{\langle 001\rangle}(T°\mathrm{K})/R_\infty^{\langle 001\rangle}\cdot(77°\mathrm{K})$ in the specimen should straightway give the absolute values of $\mu_H/\mu\langle 001\rangle_H$ provided the carrier concentration is constant. The constancy of the carrier will be discussed in the later section.

C. Evaluation of $\mu_H/\mu\langle 0\rangle_H$

Since $R_\infty^{\langle 0\rangle}(77°\mathrm{K})$ cannot be determined by simple extraporation method, $\mu_H/\mu\langle 0\rangle_H$ must be indirectly evaluated in some other way. This is possible by noting a fact that the Hall coefficient for zero field is also independent of the crystal orientations, that is to say, $R_0^{\langle 0\rangle}(T°\mathrm{K})ne$ should be equal to $R_0^{\langle 001\rangle}(T°\mathrm{K})n'e$ where n and n' are carrier concentrations of respective specimens and e is electronic charge. In the ideal case n and n' should be

equal, but actually slight difference is inevitable because of the fluctuation in impurity concentration, even if the specimens are cut from the same ingot. Fortunately, $\mu_H/\mu(T°K)$ is a rather slowly varying function of the impurity concentration. Therefore, the effect of the difference in n on μ_H/μ will be neglected here, regarding it as insignificant. Fig. 5 shows the values of $\mu_H/\mu\langle\theta\rangle_H$ determined in these manners.

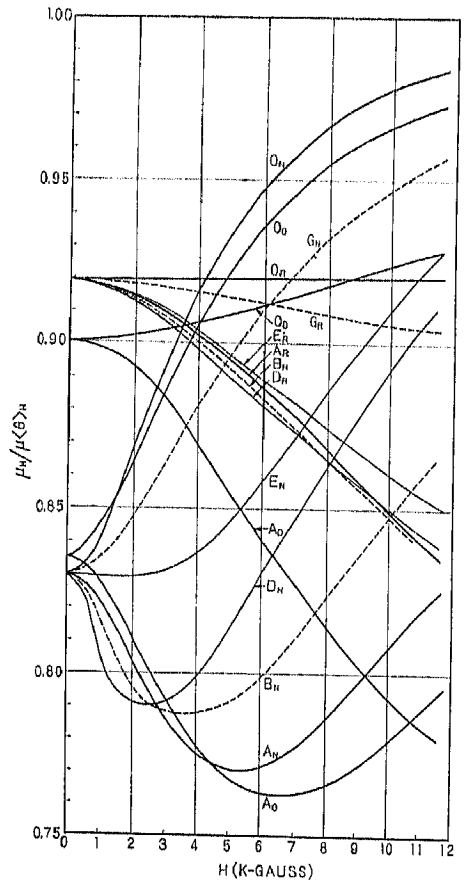


Fig. 5. Variations of $\mu_H/\mu\langle\theta\rangle_H$ at four temperatures with H.

In addition to the fact that considerably anisotropic features observed by Mason et al[5] and also by Bullis[9),10)] were reproduced here, it was explicitly examined for the first time that the values of μ_H/μ at usual magnetic field strengths and at usual exhaustion temperatures pass through a region fairly lower than unity in contrast with Morin's data[6].

As for the room temperature data, low field expansion formula,

$$R_H^{\langle\theta\rangle}=R_0(1+\xi^{\langle\theta\rangle}H^2+\eta^{\langle\theta\rangle}H^4+\zeta^{\langle\theta\rangle}H^6) \quad (3.1)$$

is valid in the whole range of measured magnetic field, and empirical values of coefficients, $\xi^{\langle\theta\rangle}$, $\eta^{\langle\theta\rangle}$ and $\zeta^{\langle\theta\rangle}$, are summarized in Table I, which also gives R_0, ρ_0 and μ_H. Theoretically calculated $\xi_a^{\langle\theta\rangle}$'s are also listed in Table I, where agreement between $\xi^{\langle\theta\rangle}$ and $\xi_a^{\langle\theta\rangle}$ seems to be satisfactory.

D. Reproductibility of $\mu_H/\mu(T°K)$

It will be desirable to check whether the assumption of constant carrier concentration irrespective of temperature is allowable or not. Though any orthodox examination is not available since variations of carrier concentration and μ_H/μ with temperatures is inseparable in the experiments, it may be worth-while to examine the reproducibility of μ_H/μ on specimens cut from different ingots which may contain different concentrations of uncotrolled centers. These uncontrolled centers which capture electrons at low temperatures may be ionized in some degree at room temperature, resulting in the increase in conduction electrons which, in turn, may cause the decrease in Hall coefficients. If observed values of $R_H^{\langle\theta\rangle}(T°K)/R_\infty(77°K)$ in each specimen were too much random, reliable determination of $\mu_H/\mu(T°K)$ may be hopeless,

Table I. Weak field expansion data at room temperature for 3 or 4 ohm-cm n-type Ge.

sample $\langle\theta\rangle_H$	A <110>	B <75°>	D <111>	E <112>	G <15°>	O <001>
R_0 (cm³/coul)	13,998	12,743	12,532	12,767	13,439	14,400
ρ_0 (ohm-cm)	4.1	3.5	3.3	3.9	4.15	3.84
μ_H (cm²/v·sec)	3,414	3,641	3,798	3,274	3,238	3,750
$\xi^{\langle\theta\rangle}$ (gauss^{-2})$\times 10^9$	−1.245	−1.298	−1.582	−1.144	−0.278	0
$\eta^{\langle\theta\rangle}$ (gauss^{-4})$\times 10^{18}$	7.112	7.952	14.00	7.676	1.762	
$\zeta^{\langle\theta\rangle}$ (gauss^{-6})$\times 10^{26}$	−2.211	−2.811	−5.366	−2.495	−0.503	
$\xi_a^{\langle\theta\rangle}$ (gauss^{-2})$\times 10^9$	−1.135	−1.275	−1.517	−1.135	−0.289	0

Table II. Reproducibility of $R_0^{\langle 001\rangle}(298°K)/R_\infty^{\langle 001\rangle}(77°K)$ for specimens cut from different ingots.

sample	$R_0(298°K)$* (cm³/coul)	$R_\infty(77°K)$ (cm³/coul)	$R_0(298°K)/R_\infty$
5-04-D	17,124	18,400	0.930
5-04-B	13,312	14,525	0.917
2-03-A	10,811	11,780	0.918
2-18-01	10,689	11,595	0.922
3-22-03	9,806	10,740	0.913
2-13-02	7,567	8,330	0.909

* $R_H^{\langle 001\rangle}(298°K)$ is field independent, and the value at 2355 gauss is taken as $R_0(298°K)$.

though larger values are favorable.

The check was carried out on several ⟨001⟩ specimens, of which results always showing fairly good reproducibility are given in Table II. From these results it may be concluded (1) that the carrier contribution from the uncontrolled centers is fortunately insignificant, or otherwise, (2) that the contribution, if any, is very similar for all ingots. If the latter were the case, the present deduction for $\mu_H/\mu(T°K)$ will be confused, and further discussion will be done in the later section.

§ 4. Theoretical Analysis

A. Model for scattering mechanism

An empirical fact of strikingly field independent $R_H(298°K)$ cannot be explained by Abeles-Meiboom's theory[2] and will suggest that slight impurity scattering is important[1] even at room temperature. In the present case, therefore, mixed scattering will be taken into account, where the relaxation time will be treated not as a power function of electron energy but as a combination of lattice and impurity scatterings. Following Herring-Vogt[8], consider an ellipsoid model for relaxation time similar to effective mass, and denote the principal value along longitudinal axis ⟨111⟩ of an energy ellipsoid as τ_l and that along transverse axis as τ_t. Then,

$$\begin{aligned} 1/\tau_l &= \sqrt{kT}/l\ (x^{1/2}+\delta^2 x^{-3/2}) \\ 1/\tau_t &= \sqrt{kT}/l(x^{1/2}+r\delta^2 x^{-3/2}) \\ x &\equiv \varepsilon/kT \end{aligned} \tag{4.1}$$

where ε is energy of electron, k Boltzmann's constant, l mean free path associated with acoustic phonon scattering, δ^2 a parameter indicating mixing rate of impurity scattering in the direction of transverse axis and r is an anisotropy parameter for pure ionized impurity scattering. It has been cleverly determined by Laff and Fan's magnetoresistance experiment[11] that r equals 1/12, while acoustic phonon scattering is isotropic. These results will be utilized here, too. According to deformation potential theory[12] and Conwell-Weisskopf's theory[13], temperature dependence of δ^2 will be approximately given by

$$\delta^2 \propto (kT)^{-3}\ . \tag{4.2}$$

If the anisotropy of impurity scattering is neglected, δ^2 corresponds to β or $6\mu_t/\mu_l$ given by Johnson-Whitesell[1]. That is to say,

$$\delta^2 \simeq 6\mu_t/\mu_l\ . \tag{4.2a}$$

B. Conductivity tensor and expression of Hall Coefficient

Now let us consider non-degenerate case and Boltzmann satistics will be assumed. From Boltzmann's equation, the conductivity tensor associated with principal axes of an ellipsoid will be given in the following form, when magnetic field $\boldsymbol{H}(H_1, H_2, H_3)$ is applied,

$$\begin{aligned} \sigma_{11} &= uK\alpha + uv^2K^2\gamma H_1^2 \\ \sigma_{22} &= uK\alpha + uv^2K^2\gamma H_2^2 \\ \sigma_{33} &= u\alpha^* + uv^2K^2\gamma H_3^2 \\ \sigma_{12} &= -uvK^2\beta H_3 + uv^2K^2\gamma H_1H_2 = \sigma_{21}(-\boldsymbol{H}) \\ \sigma_{23} &= -uvK\beta^* H_1 + uv^2K^2\gamma H_2H_3 = \sigma_{32}(-\boldsymbol{H}) \\ \sigma_{31} &= -uvK\beta^* H_2 + uv^2K^2\gamma H_3H_1 = \sigma_{13}(-\boldsymbol{H}) \\ K &\equiv m_3/m_1 = m_3/m_2 \\ u &\equiv Ne^2(l/m_3)(kT)^{-1/2}4/3\sqrt{\pi} \\ v &\equiv (e/c)(l/m_3)(kT)^{-1/2} \end{aligned} \tag{4.3}$$

where suffix 3 corresponds to longitudinal one and 1, 2 transverse, m_1, m_2 and m_3 are principal values for respective axis of effective mass tensor, K anisotropy parameter for effective mass, N carrier concentration per valley, $-e$ electronic charge. $K=20$ will be used[7] in the present analysis.

α, β etc. are integrals as shown in the following:

$$\begin{aligned}
\alpha(\delta^2,\ p;\ s) &= \int_0^\infty F^{-1}(x^2+\delta^2)(x^2+r\delta^2)x^3 e^{-x}dx \\
\alpha^*(\delta^2,\ p;\ s) &= \int_0^\infty F^{-1}(x^2+\delta^2)^2 x^3 e^{-x}dx \\
\beta(\delta^2,\ p;\ s) &= \int_0^\infty F^{-1}(x^2+r\delta^2)x^{3/2}e^{-x}dx \\
\beta^*(\delta^2,\ p;\ s) &= \int_0^\infty F^{-1}(x^2+\delta^2)x^{3/2}e^{-x}dx \\
\gamma(\delta^2,\ p;\ s) &= \int_0^\infty F^{-1}x^3 e^{-x}dx
\end{aligned} \tag{4.4}$$

where

$$\begin{aligned}
F(x,\ \delta^2,\ p;\ s) &= (x^2+\delta^2)^2(x^2+r\delta^2)+p(x^3+s\delta^2x^3) \\
p &= v^2(KH_1^2+KH_2^2+K^2H_3^2) \\
s &= (KH_1^2+KH_2^2+rK^2H_3^2)/(KH_1^2+KH_2^2+K^2H_3^2).
\end{aligned}$$

Let $\{\Sigma\}$ be conductivity tensor viewed from the observing coordinate system as chosen in Fig. 1, where magnetic field $\boldsymbol{H}$ is in z direction, current $\boldsymbol{J}$ in x and Hall voltage $\boldsymbol{V}_H$ in y direction. Then, the Hall coefficient is given by

$$R_H^{\langle 0\rangle} = \frac{1}{H}\frac{-\Sigma_{yx}}{\Sigma_{xx}\Sigma_{yy}-\Sigma_{yx}\Sigma_{xy}}. \tag{4.5}$$

Assuming four ellipsoid model possessing [111] longitudinal axes, transformation of $\{\sigma\}$ to the observing coordinate system can be performed in two steps just like Abeles-Meiboom's fashion[2]. (See Appendix.)

Several expressions interested in the present experiment will be given in the following.

(1) Isotropic quantities

Zero field Hall coefficient is independent of orientations and expressed by

$$R_0 = -\left(\frac{3\pi}{8}\right)\frac{1}{nec}\left\{\frac{2}{\sqrt{\pi}}\frac{3K(K\beta_0+2\beta_0^*)}{(2K\alpha_0+\alpha_0^*)^2}\right\} \tag{4.6}$$

where n is total carrier concentration and equal to $4N$. As for conductivity and Hall mobility,

$$\sigma_0 = (4/3)u(2K\alpha_0+\alpha_0^*) \tag{4.7}$$

$$\mu_H = |R_0\sigma_0| = |v|\frac{K(K\beta_0+2\beta_0^*)}{2K\alpha_0+\alpha_0^*}. \tag{4.8}$$

Here α_0, β_0, etc. are the values for $H=0$ or $p=0$, i.e.[14],

$$\begin{aligned}
\alpha_0 &= \int_0^\infty \frac{x^3e^{-x}dx}{x^2+\delta^2} = 3!\,\mathfrak{L}_3(\delta) \\
\alpha_0^* &= 3!\,\mathfrak{L}_3(\delta\sqrt{r}) \\
\beta_0 &= \int_0^\infty \frac{x^{3/2}e^{-x}dx}{(x^2+\delta^2)^2} = (9/2)!\,\vartheta_{3/2}(\delta) \\
\beta_0^* &= \frac{1}{1-r}(9/2)!\,\frac{1}{\delta^2}\{\mathfrak{L}_{3/2}(\delta\sqrt{r})-\mathfrak{L}_{3/2}(\delta)\}.
\end{aligned} \tag{4.9}$$

Meanwhile, infinite field Hall coefficient is also isotropic and simply given by

$$R_\infty = -\frac{1}{nec} . \tag{4.10}$$

(2) Anisotropic quantities

$$\frac{R_H^{\langle 001\rangle}}{R_0} = \frac{\dfrac{K\beta+2\beta^*}{K\beta_0+2\beta_0^*}}{\left(\dfrac{2K\alpha+\alpha^*}{2K\alpha_0+\alpha_0^*}\right)^2+\left(\dfrac{K\beta+2\beta^*}{K\beta_0+2\beta_0^*}\right)^2(\mu_H H)^2} \tag{4.11}$$

where

$$r=1/12 ,$$
$$s=(rK+2)/(K+2)=1/6$$
$$p=(1/3)\frac{(K+2)}{K}\left(\frac{2K\alpha_0+\alpha_0^*}{K\beta_0+2\beta_0^*}\right)^2(\mu_H H)^2 .$$

$$\frac{R_H^{\langle 111\rangle}}{R_0} = \frac{\dfrac{K(3\beta^{(1)}+\beta^{(2)})+8\beta^{*(2)}}{4(K\beta_0+2\beta_0^*)}}{\left\{\dfrac{3K\alpha^{(1)}+5K\alpha^{(2)}+4\alpha^{*(2)}}{4(2K\alpha_0+\alpha_0^*)}\right\}^2+\left\{\dfrac{K(3\beta^{(1)}+\beta^{(2)})+8\beta^{*(2)}}{4(K\beta_0+2\beta_0^*)}\right\}^2(\mu_H H)^2} \tag{4.12}$$

where superscripts (1) or (2) indicate that the following p and s should be substituted into α or β.

$$p=\left(\frac{2K\alpha_0+\alpha_0^*}{K\beta_0+2\beta_0^*}\right)^2(\mu_H H)^2 , \qquad s=r=1/12$$

for superscript (1), and

$$p=\left(\frac{K+8}{9K}\right)\left(\frac{2K\alpha_0+\alpha_0^*}{K\beta_0+2\beta_0^*}\right)^2(\mu_H H)^2 , \qquad s=\frac{rK+8}{K+8}=29/84$$

for superscript (2).

C. Numerical values and comparison with experiments

(1) Weak field behavior of R_H

Low field Hall coefficients may be expanded in a series with even powers of $\mu_H H$, of which form is expected from Eqs. (4.4) and (4.8) to be

$$R_H^{\langle 0\rangle}=R_0\{1+a^{\langle 0\rangle}(\mu_H H)^2+b^{\langle 0\rangle}(\mu_H H)^4-\cdots\} . \tag{4.13}$$

In order to determine the coefficients $a^{\langle 0\rangle}$ and $b^{\langle 0\rangle}$ for specific orientations, integrals in (4.11) and (4.12) should be expanded in power series of p.

$$\begin{aligned}
\alpha &= \sum_{n=0}^{\infty}(-1)^n\alpha_n p^n, \quad \alpha_n=\int_0^\infty \frac{(x^5+s\delta^2x^3)^n x^3 e^{-x}}{(x^2+\delta^2)^{2n+1}(x^2+r\delta^2)^n}dx \\
\alpha^* &= \sum_{n=0}^{\infty}(-1)^n\alpha_n^* p^n, \quad \alpha_n^*=\int_0^\infty \frac{(x^5+s\delta^2x^3)^n x^3 e^{-x}}{(x^2+\delta^2)^{2n}(x^2+r\delta^2)^{n+1}}dx \\
\beta &= \sum_{n=0}^{\infty}(-1)^n\beta_n p^n, \quad \beta_n=\int_0^\infty \frac{(x^5+s\delta^2x^3)^n x^{9/2} e^{-x}}{(x^2+\delta^2)^{2n+2}(x^2+r\delta^2)^n}dx \\
\beta^* &= \sum_{n=0}^{\infty}(-1)^n\beta_n p^n, \quad \beta^*_n=\int_0^\infty \frac{(x^5+s\delta^2x^3)^n x^{9/2} e^{-x}}{(x^2+\delta^2)^{2n+1}(x^2+r\delta^2)^{n+1}}dx .
\end{aligned} \tag{4.14}$$

Numerical values of several α_n's and β_n's are summarized in Table III. Substituting (4.14) in (4.11) and (4.12), $a^{\langle 0\rangle}$ and $b^{\langle 0\rangle}$ in (4.13) will be given by

Table III.

δ	α_0	α_0^*	$(4/3\sqrt{\pi})\beta_0$	$(4/3\sqrt{\pi})\beta_0^*$	α_1 ($s=1/6$)	α_1^* ($s=1/6$)
0	1	1	0.66667	0.66667	1	1
0.1	0.98134	0.99789	0.61617	0.63293	0.78028	0.81155
0.2	0.94825	0.99256	0.55660	0.59083	0.63141	0.68095
0.3	0.91035	0.98520	0.50127	0.54986	0.52184	0.58245
0.4	0.8712	0.97663	0.45163	0.51147	0.43545	0.50449
0.5	0.83183	0.96734	0.40760	0.47611	0.36785	0.44039
0.6	0.7938	0.95760	0.36859	0.44347	0.31310	0.38552
0.8	0.7218	0.93705	0.30338	0.38620	0.23151	0.29982
1.0	0.6564	0.91488	0.25184	0.33834	0.17435	0.23849
1.2	0.59832	0.89146	0.21074	0.29830		
1.4	0.54630	0.86982	0.17758	0.26357		
1.6	0.49998	0.84684	0.15069	0.23431		

δ	$(4/3\sqrt{\pi})\beta_1$ ($s=1/6$)	$(4/3\sqrt{\pi})\beta_1^*$ ($s=1/6$)	α_1 ($s=1/12$)	α_1^* ($s=1/12$)	$(4/3\sqrt{\pi})\beta_1$ ($s=1/12$)	$(4/3\sqrt{\pi})\beta_1^*$ ($s=1/12$)
0	1.33333	1.33333	1	1	1.33333	1.33333
0.1	0.63309	0.67312	0.77717	0.80752	0.62957	0.66842
0.2	0.43574	0.48094	0.62675	0.67464	0.43189	0.47581
0.3	0.32123	0.36383	0.51634	0.57475	0.31750	0.35858
0.4	0.24580	0.28564	0.42952	0.49594	0.24232	0.28061
0.5	0.19278	0.22943	0.36175	0.43137	0.18959	0.22471
0.6	0.15399	0.18740	0.30698	0.37626	0.15109	0.18302
0.8	0.102215	0.12968	0.22561	0.29064	0.099881	0.12595
1.0	0.069914	0.093062	0.16883	0.22963	0.067987	0.089913

δ	α_1 ($s=29/84$)	α_1^* ($s=29/84$)	$(4/3\sqrt{\pi})\beta_1$ ($s=29/84$)	$(4/3\sqrt{\pi})\beta_1^*$ ($s=29/84$)
0	1	1	1.33333	1.33333
0.1	0.78695	0.82018	0.64065	0.68318
0.2	0.64141	0.69447	0.44399	0.49191
0.3	0.53363	0.59894	0.32923	0.37507
0.4	0.44816	0.52282	0.25327	0.29640
0.5	0.38093	0.45971	0.19963	0.23953
0.6	0.32622	0.40534	0.16021	0.19679
0.8	0.24415	0.31950	0.10733	0.13766
1.0	0.18618	0.25746	0.074044	0.099811

$$a^{\langle 0\rangle}(\delta)=2A_1/S-B_1/S-1\ ,$$
$$b^{\langle 0\rangle}(\delta)=1-4A_1/S+3B_1/S-2A_2/S^2+B_2/S^2+3(A_1/S)^2-2A_1B_1/S^2 \tag{4.15}$$

where in the case of $\langle 001\rangle_H$,

$$S\equiv\frac{2K}{K+2}\left(\frac{K\beta_0+2\beta_0^*}{2K\alpha_0+\alpha_0^*}\right)^2$$
$$A_n\equiv\frac{2K\alpha_n+\alpha_n^*}{2K\alpha_0+\alpha_0^*}$$
$$B_n\equiv\frac{K\beta_n+2\beta_n^*}{K\beta_0+2\beta_0^*} \tag{4.16}$$
$$s\equiv\frac{rK+2}{K+2}=\frac{1}{6}$$

and in the case of $\langle 111\rangle_H$,

$$
\begin{aligned}
S &= \left(\frac{K\beta_0+2\beta_0^*}{2K\alpha_0+\alpha_0^*}\right)^2 \\
A_n &= \frac{1}{4(2K\alpha_0+\alpha_0^*)}\{3K\alpha_n^{(1)}+5K\alpha_n^{(2)}l^n+4\alpha_n^{*(2)}l^n\} \\
B_n &= \frac{1}{4(K\beta_0+2\beta_0^*)}\{3K\beta_n^{(1)}+K\beta_n^{(2)}l^n+8\beta_n^{*(2)}l^n\} \\
l &= \frac{K+8}{9K} = \frac{7}{45} \\
s &= r = 1/12 \qquad \text{for superscript (1)} \\
s &= \frac{rK+8}{K+8} = \frac{29}{84} \qquad \text{for superscript (2)}
\end{aligned}
\tag{4.17}
$$

Fig. 6 indicates $a^{\langle 0\rangle}(\delta)$ thus calculated.

It is a remarkable result that sign reversal of $a^{\langle 001\rangle}$ takes place at $\delta\sim 0.1$. In view of the fact that $R_H^{\langle 001\rangle}$(298°K) is independent of field strength at weak field, it is believed that δ must be nearly 0.1 at room temperature in the present crystals. Another information about δ can be obtained from (4.2a) using Conwell's μ_I for $N_I=5\times 10^{14}$ and μ_L at room temperature[13], resulting in $\delta\sim 0.14$, which agrees fairly well with the value above mentioned.

In order to compare (4.13) with (3.1), $\xi_c^{\langle 0\rangle} \equiv a^{\langle 0\rangle}/\mu_H{}^2$'s were calculated by assuming δ 0.1 at 298°K and by using $\mu_H\sim 3{,}800$ together with the theoretical curves for $a^{\langle 0\rangle}$ shown in Fig. 6. Cubic symmetry relations[5] were used to calculate $\xi_c^{\langle 0\rangle}$ except $\xi_c^{\langle 001\rangle}$ and $\xi_c^{\langle 111\rangle}$. As indicated in Table I, satisfactory agreement between $\xi^{\langle 0\rangle}$ and $\xi_c^{\langle 0\rangle}$ was obtained. These features will predict weak field behaviors of Hall coefficients for crystals of any resistivity or at any temperature not actually measured, by sliding δ with the relations of (4.2) and (4,2a).

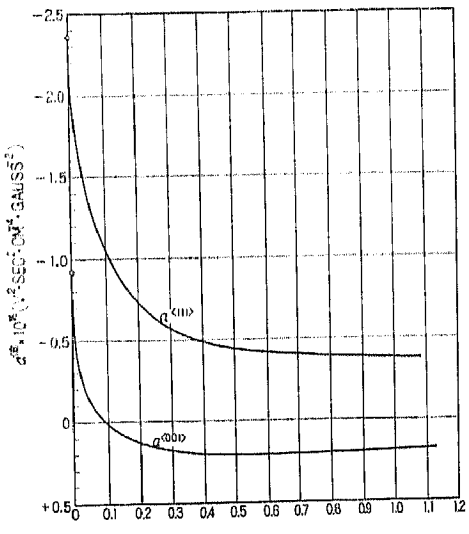


Fig. 6. Weak field expansion coefficient $a^{\langle 0\rangle}(\delta)$.

(2) Temperature dependence of R_0/R_∞

From (4.6) and (4.10) R_0/R_∞ or μ_H/μ at zero field is easily obtained.

$$
\begin{aligned}
\frac{R_0}{R_\infty} &= \frac{3\pi}{8}\frac{3K(K+2)}{(2K+1)^2}I(\delta) \\
I(\delta) &= \frac{2}{\sqrt{\pi}}\frac{(2K+1)^2}{K+2}\frac{(K\beta_0+2\beta_0^*)}{(2K\alpha_0+\alpha_0^*)^2}
\end{aligned}
\tag{4.18}
$$

Fig. 7 shows the behavior of $I(\delta)$, where solid curve is calculated with $K=20$ and $r=1/12$, while two broken curves correspond (1) to the one in which $K=15$ and $r=1/12$ are assumed, and (2) to the one in which $K=20$ and $r=1$, respectively. It is found that

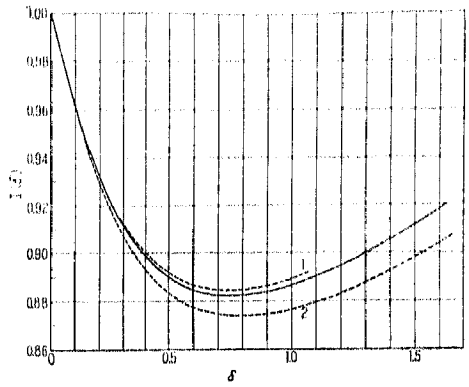


Fig. 7. $I(\delta)$ vs δ. Solid curve; $K=20$, $r=1/12$.
Broken curve 1; $K=15$, $r=1/12$.
Broken curve 2; $K=20$, $r=1$.

isotropic impurity scattering depresses the minimum of $I(\delta)$ to some extent, while mass ratio does not too much affect $I(\delta)$. When $\delta=0$, (4.18) coincides with Abeles-Meiboom or Shibuya's formula[2),3)].

For the purpose of comparison with experimental data, it will be convenient to draw the R_0/R_∞ curve not against δ but against temperature. From (4.2) and previous inference on δ, let us now assume

$$\delta(T)=0.1(300/T)^{3/2}. \qquad (4.19)$$

Substituting (4.19) in (4.18), R_0/R_∞ will become a function of temperature. The resultant R_0/R_∞ vs T curve is illustrated in Fig. 8. It is seen that R_0/R_∞ has a minimum of 0.82 near liquid nitrogen temperature in the present case. Meanwhile, under constant mean free time assumption investigated by Bullis and Krag[6] the minimum of R_0/R_∞ was 0.79. Of course, better agreement with experiments can be obtained in the present case, though exact experimental determination of R_0's is somewhat difficult at 77°K and 90°K. The discrepancy at zero field may be a fault of the constant mean free time approximation.

In Fig. 8 two other R_0/R_∞ vs T curves, in which (1) δ is chosen by Conwell-Weisskopf's formula or (2) isotropic impurity scattering is assumed, are also presented. It seems theoretical R_0/R_∞'s have qualitatively explained the behavior of experimental data. However, a tendency cannot be overlooked that the observed values deviate upwards from the theoretical curves, especially at higher temperatures, where no possibility depressing the observed points is expected. As will be discussed in the later section, this might suggest that some correction such as optical phonon scattering is necessary. Remarks on the curve 3 in Fig. 8 will be given in the next section.

§ 5. Discussions

A. Constancy of carriers

Applying Fermi satistics to a semiconductor containing 5×10^{14} donors, N_d, per cm^3 of which energy level, E_d, lies at 0.01 eV below the conduction band edge, the rate of exhaustion can be easily determined at any temperature. This is the ideal case in the pre-

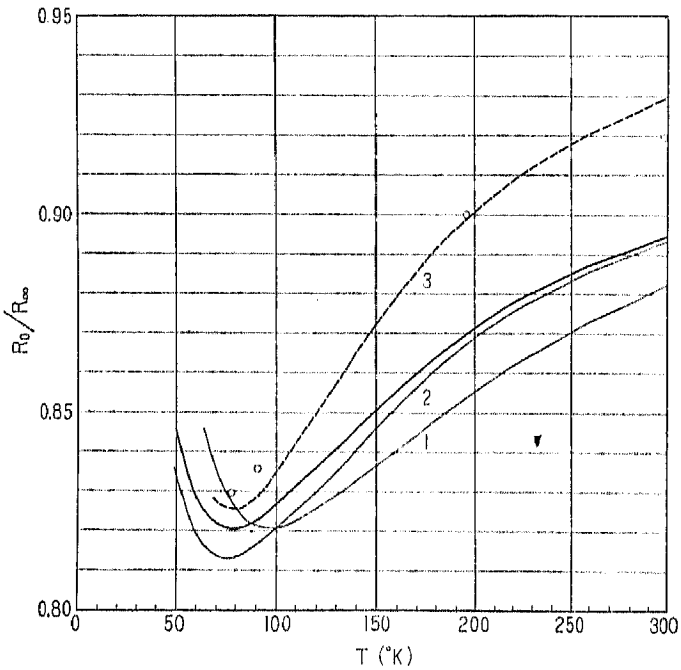


Fig. 8. R_0/R_∞ vs T. Open circles are experimental. Solid curve; calculated with $\delta=0.1\cdot(300/T)^{3/2}$, $K=20$, $r=1/12$. Thin curve 1; $\delta=0.14(300/T)^{3/2}$, derived from Conwell's μ_I is used instead of above δ. Thin curve 2; $r=1$ or isotropic impurity scattering is assumed, leaving δ and K unchanged. Broken curve 3; modified by optical phonon scattering.

Table IV. Rate of exhaustion and corresponding Fermi energy in a semiconductor, in which $N_d=5\times10^{14}$ cm^{-3}, $E_d=0.01$ eV and $(m^*/m)^{3/2}=0.4$.

	77°K	90°K	195°K	298°K
$N_d/n(T°\mathrm{K})$	1.0017	1.0011	1.0002	1.00006
$n(T°\mathrm{K})/n(77°\mathrm{K})$	1.0000	1.0006	1.0015	1.0016
$\zeta(T°\mathrm{K})$ eV	−0.052	−0.063	−0.156	−0.254

sent specimens. Table IV shows the ratio $N_d/n(T°\mathrm{K})$, $n(T°\mathrm{K})/n(77°\mathrm{K})$ and Fermi energy $\zeta(T°\mathrm{K})$. It is seen that the assumption of constant carriers is well assured in the ideal case.

However, actual specimens may naturally contain some uncontrolled imperfections, from which ionization is possible at higher temperatures. Then, let us consider the maximum concentration of these uncontrolled centers. One clue will be to measure the mean lifetime of injected minority carriers. In most crystals used here the lifetime by means of photoconductivity decay method was about 200 μsec at room temperature. According to Burton et al[15] the value of 200 μsec is their highest limit for 3 ohm-cm n-type germanium, where the concentration of the unknown recombination centers, N_r, may be estimated to be less than 1×10^{13} per cm^3. Though it is not clear what the sources of N_r are, probably they consist of crystal imperfections such as dislocations or vacancies and chemical impurities. As a matter of fact main centers concerned with the carrier contribution are those having energy levels near or between Fermi levels at liquid nitrogen and room temperatures. Therefore, dislocations, which have been proved to have an acceptor level about 0.2 eV below the conduction band[16], must be considered in the first place, since they are substantially empty at room temperature but can occupy electrons at low temperatures. According to the reciprocal law between lifetime and dislocation density[17], the lifetime of nearly 500 μsec, which is two or three times was large as the observed one, will be expected at room temperature using measured dislocation density of about 5×10^3 per cm^2 or less in our crystals. Thus it would appear that dislocations are an important factor among N_r concerned with the carrier contribution, even if Read's theory has concluded[18] that only relatively few of dislocation sites will be filled at low temperatures because of space charge effect. The remaining fraction of N_r might be due to chemical impurities, copper for example, and frozen-in vacancies, though their solubility[19] is rather small. These centers possess their energy levels near or below the middle of the gap[20] and do not much affect the number of carriers.

In consequence, the temperature effect of uncontrolled centers on number of carriers would be fortunately insignificant, and the resultant undesirable shift of $R_H(T°\mathrm{K})$ due to carrier change should be at most within 2 percent even in the worst case.

B. Thermomagnetic effect

Since the Ettinghausen effect cannot be excluded in the experiment, it will be necessary to confirm negligible contribution of the effect to Hall voltage. Ettinghausen effect, entirely the same with Hall effect, can be written in the form

$$\Delta T=PIH/d \tag{5.1}$$

where ΔT, P, I, and d are temperature difference across the specimen, Ettinghausen coefficient, total current and thickness of the specimen respectively. As for P, Mette et al[21] have obtained the value of nearly 10^{-6} deg. cm. amp.$^{-1}$ gauss^{-1} for 3 ohm-cm n-type germanium at 300°K. Using this result with the present experimental conditions of $I\sim0.5$ mA and $d\sim1$ mm, (5.1) gives $\Delta T\sim5\times10^{-5}$ deg. at $H=10^4$ gauss. Considering the Seebeck effect of which order is 1 mV per deg. carried out by Geballe and Hull,[22] the resultant electromotive force should be approximately equal to 0.05 μV. Consequently, this is quite negligible as compared with Hall voltage of about 10 mV at 10^4 gauss, and it may be concluded that even phonon-drag effect at low temperatures is out of the question.

C. Optical phenon scattering

Plotting the observed values of R_0/R_∞ against temperature, it seems that they are some-

what lager than the theoretical ones as already shown in Fig. 8. Furthermore, there may be a possibility that $R_0/R_\infty(298^\circ\mathrm{K})$ shifts within 2 per-cent toward higher value, in view of the doubt in carrier constancy. Though the discrepancy is not so definite that the theoretical analysis may be radically modified, it is yet suggested that some correction is desirable.

One resonable approach will be to consider the effect of optical phonon scattering, which has bee used to explain a fact that the temperature dependence of conduction electron mobility does not obey the exact $-3/2$ power law but rather follows -1.66 power law[6] in very pure crystals. In order to understand this behavior, Herring[23] has proposed to take account of intervalley scattering, while Weinreich et al[24] have insisted that the effect should be ascribed not to intervalley scattering but to optical phonon scattering with their experiments on acousto-electric effect.

Then, a trial calculation was carried out by adding a term $1/\tau_0$ due to optical phonon to Eq. (4.1). Here, $1/\tau_0$ is given by

$$1/\tau_0=\frac{\sqrt{kT}}{l}\begin{cases} C\dfrac{\gamma}{e^\gamma-1}\sqrt{x+\gamma}+C\dfrac{\gamma}{1-e^{-\gamma}}\sqrt{x-\gamma} & x>\gamma \\ C\dfrac{\gamma}{e^\gamma-1}\sqrt{x+\gamma} & x<\gamma, \end{cases} \tag{5.2}$$

where $\gamma\equiv\theta_0/T$. Using $\theta_0\sim 432^\circ$K obtained by Brockhouse et al[25] by means of neutron diffraction, and by assuming $C=0.125$ which seems to be acceptable, broken curve 3 shown in Fig. 8 has been calculated, resulting in better agreement with the observed values.

§ 6. Conclusions and Acknowledgments

Anisotropic values of μ_H/μ appearing in Hall coefficients in n-type germanium have been precisely observed at four specifield temperatures and theoretically discussed under ellipsoid model and mixed scattering. Experimental results have elucidated the importance of slight impurity scattering. It has been also clarified that μ_H/μ's at usual magnetic field strengths are considerably anisotropic and almost always smaller than unity. These features are well explained by the theoretical analysis.

In case μ_H/μ is assumed to be $3\pi/8$ in ordinary Hall measurements, it is rather recommended that a value of 0.9, in place of $3\pi/8$, is used in non-degenerate n-type germa nium. For further arguments, the following facts should be remembered even if number of carriers remains unchanged:

(1) R_0 decreases at first with decreasing temperature, reaches a minimum, and then begins to rise again.

(2) Unless θ is very small, $R_H^{[0]}$ generally decreases at first with increasing field, reaches a minimum, and then increases toward its infinite field value.

(3) $R_H^{\langle 001\rangle}$ approaches its infinite field value in most straightforward way without initial decreasing.

Careful experiments of this kind may be utilized to detect the small mixing rate of ionized impurity scattering into lattice scattering or may be helpful to determine the symmetry property of energy band structure in a semiconductor. These knowledges will be also useful to apply the Hall effect to some devices.

Finally, the authors wish to acknowledg the support of Dr. H. Muraoka and Mr. T. Abe of Toshiba Transistor Plant who prepared and oriented the crystal. They are also greateful to Dr. M. Shibuya of the Electrotechnical Laboratory for many helpful suggestions and discussions. Mr. T. Kikuchi and Miss E. Komiyama assisted with many of the numerical calculations and the experimental measurements.

Appendix I. Transformation of $\{\sigma\}$ into $\{\Sigma\}$

(1) $\{S\}$, Conductivity tensor in respect to crystal axis system

The transformation must be performed in two steps. First, the components of $\{\sigma\}^{(i)}$, the conductivity tensor viewed from the i-th valley system (i), are transformed into the crystal

axis system (c), while $\boldsymbol{H}^{(i)}$ in $\{\sigma\}^{(i)}$ are fixed as parameters. Here $\boldsymbol{H}^{(i)}$ is the set of the components of magnetic field $\boldsymbol{H}$ viewed from the i-th valley system. Let the transformation tensor be $\{\mathrm{A}\}^{(i)}$, defined for a vector $\boldsymbol{v}$ as

$$\boldsymbol{v}^{(i)}=\{\mathrm{A}\}^{(i)}\cdot\boldsymbol{v}^{(a)}\ . \tag{A.1}$$

Then we obtain

$$\{\mathrm{S}'\}=\sum_i\{\mathrm{A}\}^{(i)-1}\cdot\{\sigma(\boldsymbol{H}^{(i)})\}^{(i)}\cdot\{\mathrm{A}\}^{(i)}\ . \tag{A.2}$$

Secondly, $\boldsymbol{H}^{(i)}$'s in $\{\mathrm{S}'\}$ are replaced by the original magnetic field referred to the new system. This can be performed by substituting

$$\boldsymbol{H}^{(i)}=\{\mathrm{A}\}^{(i)}\cdot\boldsymbol{H}^{(a)} \tag{A.3}$$

in (A.2).

Table A.I.

$\{\mathrm{A}\}^{(1)}$ $\langle 111\rangle$, $\langle\bar{1}\bar{1}1\rangle$			$\{\mathrm{A}\}^{(2)}$ $\langle\bar{1}11\rangle$, $\langle 1\bar{1}1\rangle$			$\{\mathrm{A}\}^{(3)}$ $\langle\bar{1}\bar{1}1\rangle$, $\langle 11\bar{1}\rangle$			$\{\mathrm{A}\}^{(4)}$ $\langle\bar{1}11\rangle$, $\langle 1\bar{1}\bar{1}\rangle$		
$-1/\sqrt{2}$	$1/\sqrt{2}$	0	$1/\sqrt{2}$	$1/\sqrt{2}$	0	$1/\sqrt{2}$	$-1/\sqrt{2}$	0	$-1/\sqrt{2}$	$-1/\sqrt{2}$	0
$-1/\sqrt{6}$	$-1/\sqrt{6}$	$2/\sqrt{6}$	$-1/\sqrt{6}$	$1/\sqrt{6}$	$2/\sqrt{6}$	$1/\sqrt{6}$	$1/\sqrt{6}$	$2/\sqrt{6}$	$1/\sqrt{6}$	$-1/\sqrt{6}$	$2/\sqrt{6}$
$1/\sqrt{3}$	$1/\sqrt{3}$	$1/\sqrt{3}$	$1/\sqrt{3}$	$-1/\sqrt{3}$	$1/\sqrt{3}$	$-1/\sqrt{3}$	$-1/\sqrt{3}$	$1/\sqrt{3}$	$-1/\sqrt{3}$	$1/\sqrt{3}$	$1/\sqrt{3}$

In n-type germanium, $\{\mathrm{A}\}^{(i)}$'s are given in Table A.I. Thus we obtain, denoting the field components referred to the crystal axis system as H_1, H_2, H_3,

$$\begin{aligned}
S_{11}&=u\sum_{i=1}^{4}[(1/3)\{2K\alpha(p^{(i)},\ s^{(i)})+\alpha^*(p^{(i)},\ s^{(i)})\}+v^2K^2H_1^2\gamma(p^{(i)},\ s^{(i)})\\
S_{12}&=u\sum_{i=1}^{4}[(-1)^i(1/3)\{K\alpha(p^{(i)},\ s^{(i)})-\alpha^*(p^{(i)},\ s^{(i)})\\
&\quad-(1/3)(vK\{Kq^{(i)}\beta(p^{(i)},\ s^{(i)})+q^{*(i)}\beta^*(p^{(i)},\ s^{(i)})+v^2K^2H_1H_2\gamma(p^{(i)},\ s^{(i)})\\
S_{12}(\boldsymbol{H})&=S_{21}(-\boldsymbol{H})\ .
\end{aligned} \tag{A.4}$$

Here we must substitute the following relations into $p^{(i)}$'s, $s^{(i)}$'s, $q^{(i)}$'s and $q^{*(i)}$'s according to Eq. (A.3).

$$\begin{aligned}
p_1&=(1/3)v^2K\{(K+2)(H_1^2+H_2^2+H_3^2)+2(K-1)(H_1H_2+H_2H_3+H_3H_1)\}\\
p_2&=(1/3)v^2K\{(K+2)(H_1^2+H_2^2+H_3^2)+2(K-1)(-H_1H_2-H_2H_3+H_3H_1)\}\\
p_3&=(1/3)v^2K\{(K+2)(H_1^2+H_2^2+H_3^2)+2(K-1)(H_1H_2-H_2H_3-H_3H_1)\}\\
p_4&=(1/3)v^2K\{(K+2)(H_1^2+H_2^2+H_3^2)+2(K-1)(-H_1H_2+H_2H_3-H_3H_1)\}\\
s_1p_1&=(1/3)v^2K\{(rK+2)(H_1^2+H_2^2+H_3^2)+2(rK-1)(H_1H_2+H_2H_3+H_3H_1)\}\\
s_2p_2&=(1/3)v^2K\{(rK+2)(H_1^2+H_2^2+H_3^2)+2(rK-1)(-H_1H_2-H_2H_3+H_3H_1)\}\\
s_3p_3&=(1/3)v^2K\{(rK+2)(H_1^2+H_2^2+H_3^2)+2(rK-1)(H_1H_2-H_2H_3-H_3H_1)\}\\
s_4P_4&=(1/3)v^2K\{(rK+2)(H_1^2+H_2^2+H_3^2)+2(rK-1)(-H_1H_2+H_2H_3-H_3H_1)\}
\end{aligned} \tag{A.5}$$

$$\begin{aligned}
q_1&=H_1+H_2+H_3 & q_1^*&=-H_1-H_2+2H_3\\
q_2&=H_1-H_2+H_3 & q_2^*&=-H_1+H_2+2H_3\\
q_3&=-H_1-H_2+H_3 & q_3^*&=H_1+H_2+2H_3\\
q_4&=-H_1+H_2+H_3 & q_4^*&=H_1-H_2+2H_3\ .
\end{aligned} \tag{A.6}$$

The other components of S_{jk} are obtained by cyclic permutation of indices of H_i in Eqs. (A.4), (A.5) and (A.6).

(2) $\{\Sigma\}$, Conductivity tensor in respect to observing system

The transformation can be performed in the same way as in (1). Now let us define the tensor $\{G\}$ as follows

$$\boldsymbol{v}^{(a)}=\{G\}\cdot\boldsymbol{v}^{(0)}\ , \tag{A.7}$$

or

$$\begin{aligned} J//\langle G_{11},\ G_{12},\ G_{13}\rangle \\ V_H//\langle G_{21},\ G_{22},\ G_{23}\rangle \\ H//\langle G_{31},\ G_{32},\ G_{33}\rangle\ . \end{aligned} \tag{A.8}$$

Then we obtain

$$\{\Sigma'\}=\{G\}\cdot\{S\}\cdot\{G\}^{-1}\ . \tag{A.9}$$

Subsequently, $\boldsymbol{H}$ in (A.9) must be replaced as

$$H_1=G_{31}H,\ H_2=G_{32}H,\ H_3=G_{33}H\ . \tag{A.10}$$

In the present experimental situation,

$$\{G\}=\begin{vmatrix} -1/\sqrt{2} & 1/\sqrt{2} & 0 \\ -(1/\sqrt{2})\cos\theta & -(1/\sqrt{2})\cos\theta & \sin\theta \\ (1/\sqrt{2})\sin\theta & (1/\sqrt{2})\sin\theta & \cos\theta \end{vmatrix} \tag{A.11}$$

in accordance with Fig. 1.

For $\langle 001\rangle_H$ case, for instance, we obtain from (A.9), (A.10) and (A.11),

$$\begin{aligned} \Sigma_{11}&=(1/2)(S_{11}+S_{22}-S_{12}-S_{21}) \\ \Sigma_{22}&=(1/2)(S_{11}+S_{22}+S_{12}+S_{21}) \\ \Sigma_{12}&=(1/2)(S_{11}-S_{22}+S_{12}-S_{21}) \\ \Sigma_{21}&=(1/2)(S_{11}-S_{22}-S_{12}+S_{21})\ , \end{aligned} \tag{A.12}$$

where

$$\begin{aligned} S_{11}&=S_{22}=(4/3)u\{2K\alpha(p,\ s)+\alpha^*(p,\ s)\} \\ S_{12}&=-S_{21}=-(4/3)uvK\{K\beta(p,\ s)+2\beta^*(p,\ s)\}H\ . \end{aligned} \tag{A.13}$$

In this case, following parameters have been used which are derived from (A.5) and (A.6) together with (A.10) and (A.11).

$$\begin{aligned} p&\equiv p_1=\cdots=p_4=(1/3)v^2K(K+2)H^2 \\ s&\equiv s_1=\cdots=s_4=(rK+2)/(K+2) \\ q_1&=\cdots=q_4=H \\ q_1^*&=\cdots=q_4^*=2H\ . \end{aligned} \tag{A.14}$$

We obtain $R_H^{\langle 001\rangle}$ by substituting (A.12) and (A.13) into (4.5), and then R_0 by taking $p=0$, from which (4.11) can be derived. (4.12) is also obtained in the similar way.

Appendix II. Condition for weak field expansion

Expansion of α etc. in respect to p is performed by rewriting the denominator of integrand in the following form,

$$\frac{1}{(x^2+\delta^2)^2(x^2+r\delta^2)+px^2(x^2+s\delta^2)}=\frac{1}{(x^2+\delta^2)^2(x^2+r\delta^2)}\left[1-\sum_{n=1}^{\infty}\left\{\frac{x^2(x^2+s\delta^2)}{(x^2+\delta^2)^2(x^2+r\delta^2)}\right\}^n p^n\right]\ . \tag{A.15}$$

In the integration of every term the expansion is meaningless unless

$$\alpha_n p^n > \alpha_{n+1} p^{n+1}, \quad \text{etc.} \tag{A.16}$$

Therefore the condition will be sought here. Since $s<1$, the sufficient condition will be given by

$$\frac{px^n(x^2+s\delta^2)}{(x^2+\delta^2)^n(x^2+r\delta^2)} < \frac{px^n}{(x^2+\delta^2)(x^2+r\delta^2)} < 1 \qquad \text{for } 0<x<\infty, \tag{A.17}$$

or

$$(x^2+\delta^2)(x^2+r\delta^2)-px^n=\delta^4\{(\xi^2+1)(\xi^2+r)-\varphi\xi^n\}=\delta^4 f(\xi)>0, \tag{A.18}$$

where

$$\xi = x/\delta, \qquad \varphi = p/\delta.$$

From

$$f'(\xi)=\xi\{4\xi^2-3\varphi\xi+2(1+r)\}=0, \quad \text{and} \quad r=1/12$$

the minimum value of $f(\xi)$ is given at

$$\xi_0=(3\varphi+\sqrt{9\varphi^2-(104/3)})/8. \tag{A.19}$$

Numerical evaluation of $f_{\min}>0$ must result in the condition

$$\varphi = p/\delta < 0.2. \tag{A.20}$$

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